LinkedIn Job Market Analysis Report

A data-driven exploration of talent competition, hiring trends, experience demand, and sector activity across 949 job postings

Dataset	LinkedIn Job Postings
Total Records	949 Rows
Total Columns	16 Columns
Key Fields	Title, Company, Sector
Time Dimension	publishedAt (Date)
Geography	City, State
Source Platform	LinkedIn

Prepared by:
[PATHAN AFRIDH KHAN] | Data Analyst
Tools Used: SQL /,Excel
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PART A: Basic Understanding

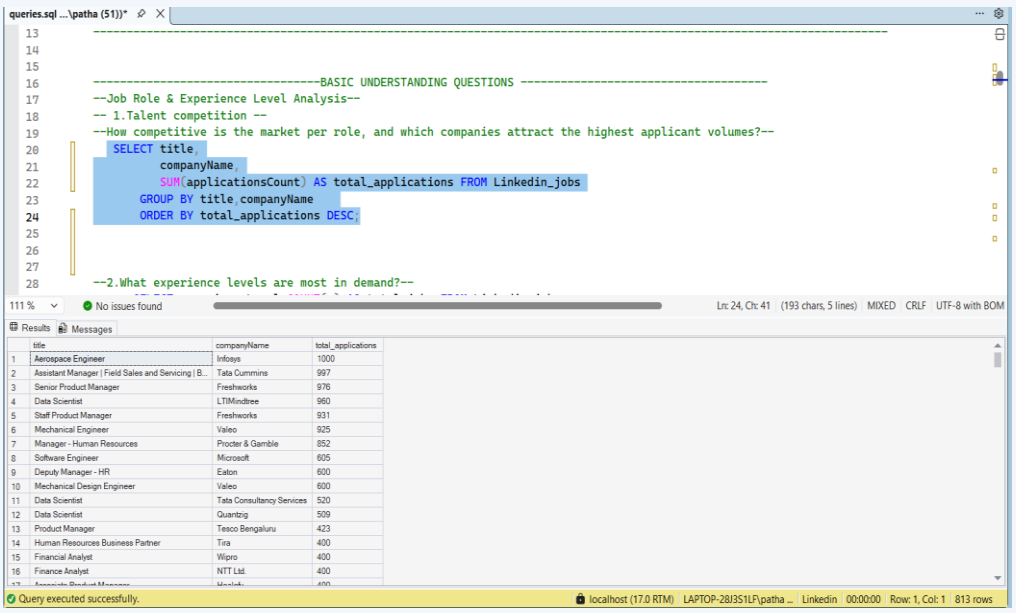
Q1. Talent Competition – Market Competitiveness & Applicant Volumes

This analysis measures how competitive each role is by examining applicant volumes across job titles and companies. High applicant counts signal roles where candidates face stiff competition, helping both job seekers and recruiters benchmark expectations.

SQL Query Used:

```
SELECT title,
       companyName,
       SUM(applicationsCount) AS total_applications FROM Linkedin_jobs
    GROUP BY title,companyName
    ORDER BY total_applications DESC;
```

Output / Visualization:



	title	companyName	total_applications
1	Aerospace Engineer	Infosys	1000
2	Assistant Manager | Field Sales and Servicing | B...	Tata Cummins	997
3	Senior Product Manager	Freshworks	976
4	Data Scientist	LTIMindtree	960
5	Staff Product Manager	Freshworks	931
6	Mechanical Engineer	Valeo	925
7	Manager - Human Resources	Procter & Gamble	852
8	Software Engineer	Microsoft	605
9	Deputy Manager - HR	Eaton	600
10	Mechanical Design Engineer	Valeo	600
11	Data Scientist	Tata Consultancy Services	520
12	Data Scientist	Quantzig	509
13	Product Manager	Tesco Bengaluru	423
14	Human Resources Business Partner	Tira	400
15	Financial Analyst	Wipro	400
16	Finance Analyst	NTT Ltd.	400

Q2. Experience Levels Most In Demand

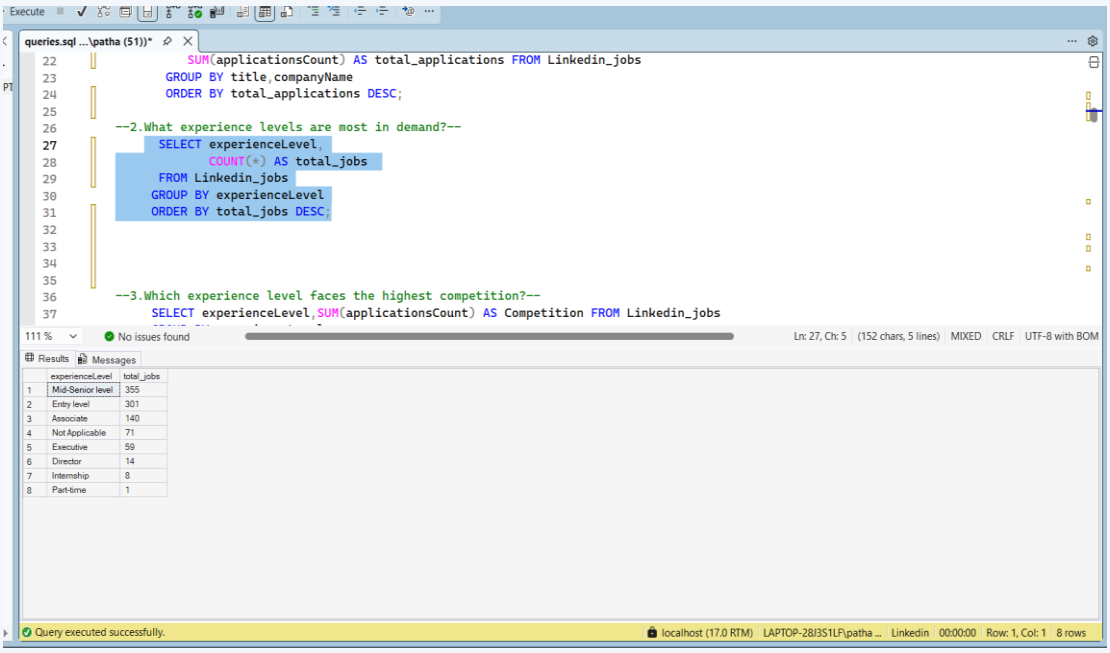
Understanding which experience levels employers are targeting the most reveals talent pipeline priorities and helps candidates position themselves accordingly in a competitive market.

SQL Query Used:

```
    SELECT experienceLevel,
```

```
        COUNT(*) AS total_jobs
 FROM Linkedin_jobs
GROUP BY experienceLevel
ORDER BY total_jobs DESC;
```

Output / Visualization:



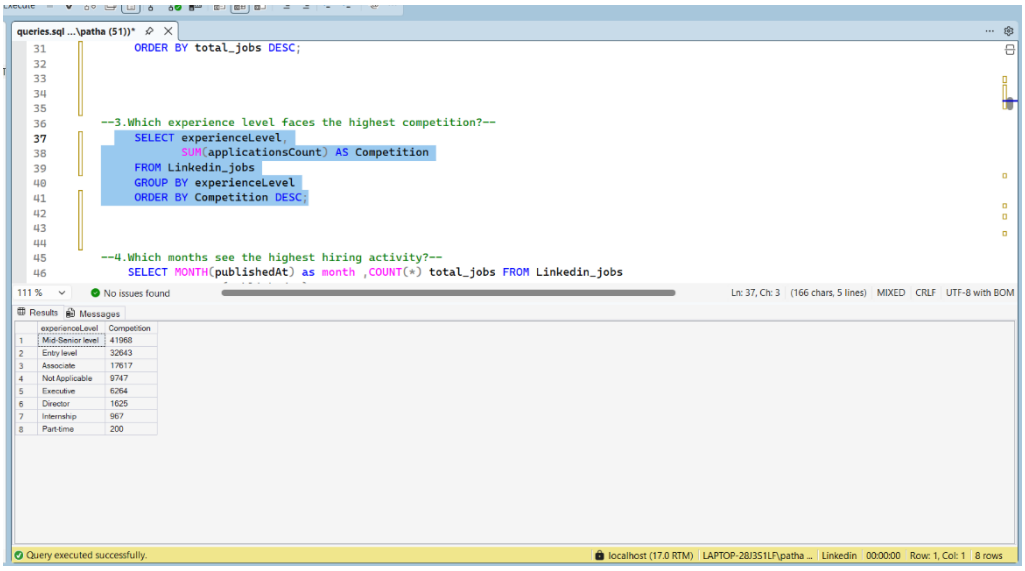
Q3. Which Experience Level Faces the Highest Competition?

By dividing total applicants by the number of postings at each experience level, we derive the applicants-per-posting ratio — a direct proxy for competition intensity faced by candidates at each career stage.

SQL Query Used:

```
SELECT experienceLevel,
       SUM(applicationsCount) AS Competition
FROM Linkedin_jobs
GROUP BY experienceLevel
ORDER BY Competition DESC;
```

Output / Visualization:



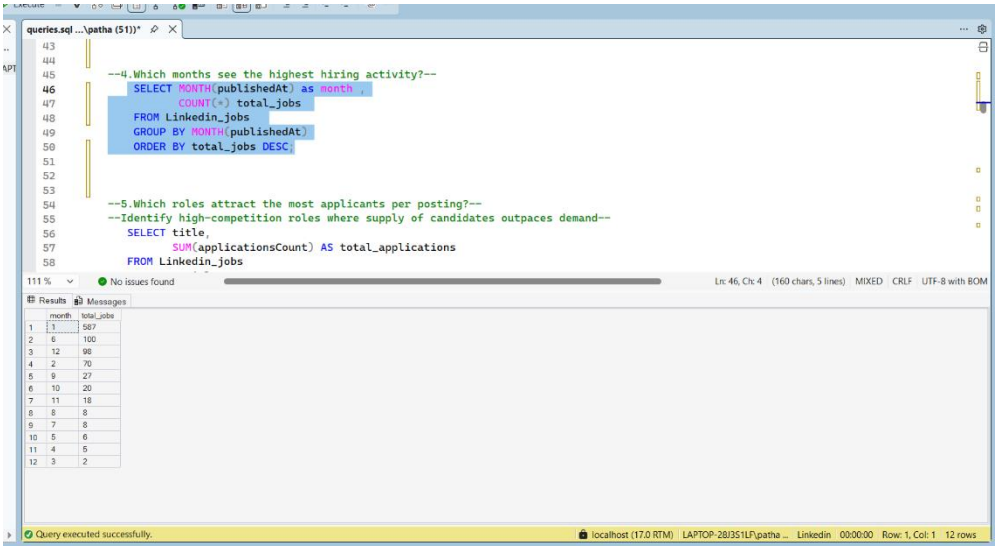
Q4. Which Months See the Highest Hiring Activity?

Seasonal patterns in job postings reveal when employers ramp up recruitment. Understanding these cycles gives candidates a strategic advantage in timing their job search efforts.

SQL Query Used:

```
SELECT MONTH(publishedAt) as month ,
       COUNT(*) total_jobs
FROM Linkedin_jobs
GROUP BY MONTH(publishedAt)
ORDER BY total_jobs DESC;
```

Output / Visualization:



Key Findings:

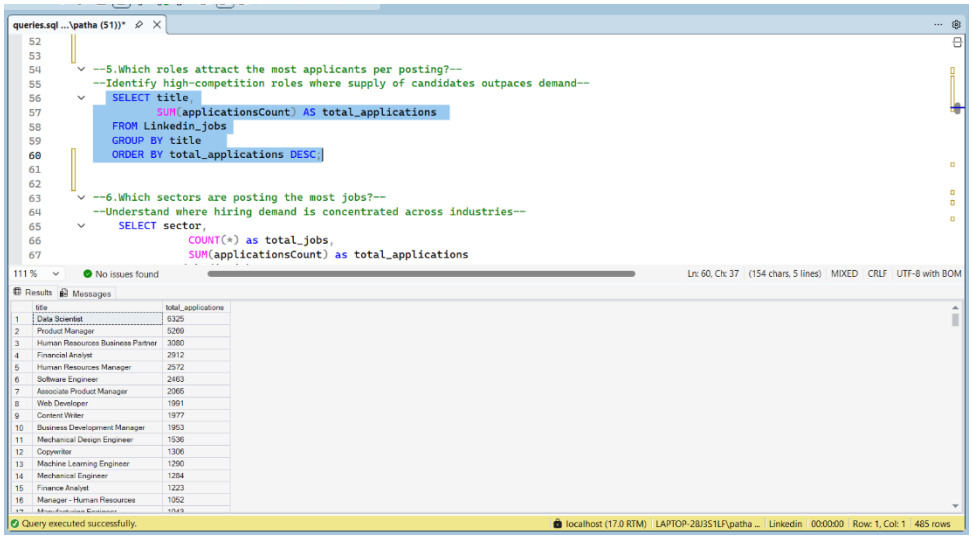
Q5. Which Roles Attract the Most Applicants Per Posting?

Roles with the highest applicant-to-posting ratios are the most fought-over positions in the market. This metric highlights where candidates should prepare the most to stand out.

SQL Query Used:

```
SELECT title,
       SUM(applicationsCount) AS total_applications
FROM Linkedin_jobs
GROUP BY title
ORDER BY total_applications DESC;
```

Output / Visualization:



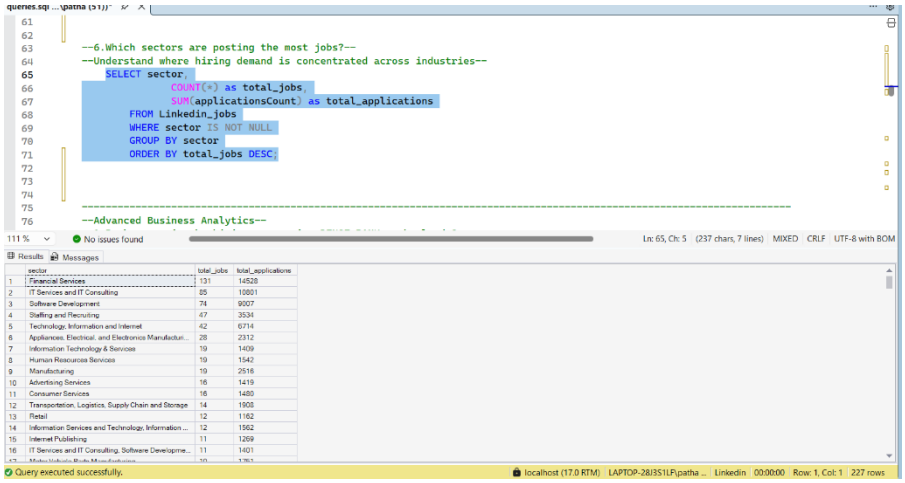
Q6. Which Sectors Are Posting the Most Jobs?

Sector-level posting volume reflects where industry growth and investment are concentrated. This helps candidates and analysts identify thriving industries versus those seeing hiring slowdowns.

SQL Query Used:

```
SELECT sector,
          COUNT(*) as total_jobs,
          SUM(applicationsCount) as total_applications
    FROM Linkedin_jobs
    WHERE sector IS NOT NULL
    GROUP BY sector
    ORDER BY total_jobs DESC;
```

Output / Visualization:



```sql
--6.Which sectors are posting the most jobs?--
--Understand where hiring demand is concentrated across industries--
    SELECT sector,
           COUNT(*) as total_jobs,
           SUM(applicationsCount) as total_applications
    FROM Linkedin_jobs
    WHERE sector IS NOT NULL
    GROUP BY sector
    ORDER BY total_jobs DESC;


--------------------------------------------------------------------------------------------------------------------
--Advanced Business Analytics--
```

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Results Messages

	sector	total_jobs	total_applications
1	Financial Services	131	14528
2	IT Services and IT Consulting	85	10801
3	Software Development	74	9007
4	Staffing and Recruiting	47	3534
5	Technology, Information and Internet	42	6714
6	Appliances, Electrical, and Electronics Manufactur...	28	2312
7	Information Technology & Services	19	1409
8	Human Resources Services	19	1542
9	Manufacturing	19	2516
10	Advertising Services	16	1419
11	Consumer Services	16	1480
12	Transportation, Logistics, Supply Chain and Storage	14	1908
13	Retail	12	1162
14	Information Services and Technology, Information ...	12	1562
15	Internet Publishing	11	1269
16	IT Services and IT Consulting, Software Developme...	11	1401

Query executed successfully. localhost (17.0 RTM) LAPTOP-28I3S1LP\patha ... Linkedin 00:00:00 Row: 1, Col: 1 227 rows

PART B: Advanced Business Analytics

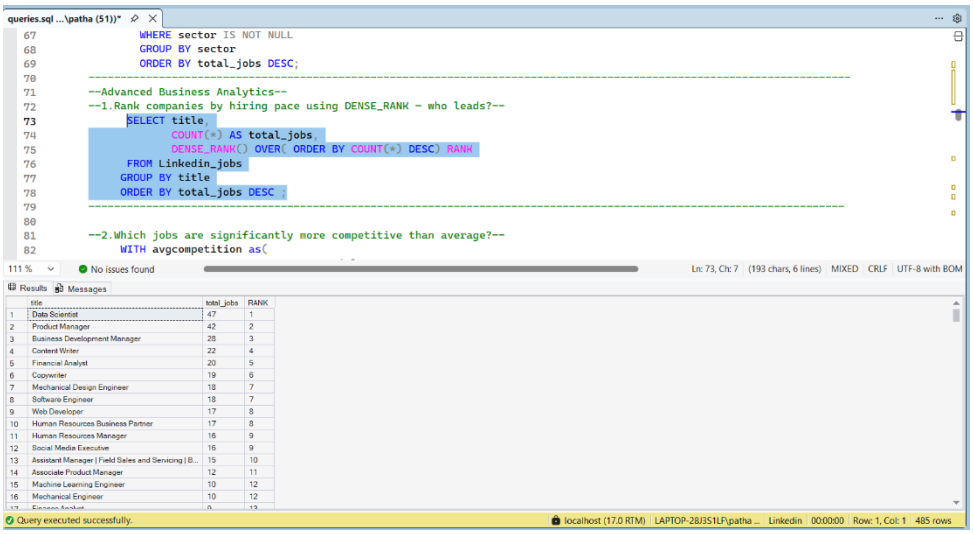
Q1. Companies Ranked by Hiring Pace Using DENSE_RANK

Using SQL's DENSE_RANK window function, companies are ranked by their total job postings and applicant volumes. This reveals which organizations are leading the market in terms of raw hiring velocity.

SQL Query Used:

```
SELECT title,
       COUNT(*) AS total_jobs,
       DENSE_RANK() OVER( ORDER BY COUNT(*) DESC) RANK
FROM Linkedin_jobs
GROUP BY title
ORDER BY total_jobs DESC ;
```

Output / Visualization:



Q2. Which Jobs Are Significantly More Competitive Than Average?

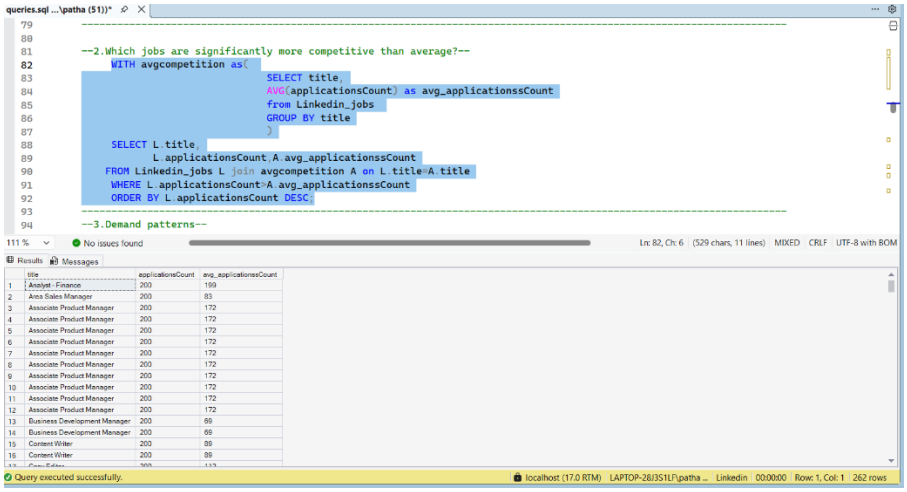
By comparing each job's applicant count against the dataset average, we flag roles where competition is statistically above the norm — critical intelligence for both candidates and talent teams.

SQL Query Used:

```
WITH avgcompetition as(
                        SELECT title,
                        AVG(applicationsCount) as avg_applicationssCount
                        from Linkedin_jobs
                        GROUP BY title
                        )
SELECT L.title,
```

```
        L.applicationsCount,A.avg_applicationssCount
FROM Linkedin_jobs L join avgcompetition A on L.title=A.title
 WHERE L.applicationsCount>A.avg_applicationssCount
 ORDER BY L.applicationsCount DESC;
```

Output / Visualization:



Q3. Demand Patterns – Which Sectors, Cities, and Roles Are Surging or Declining?

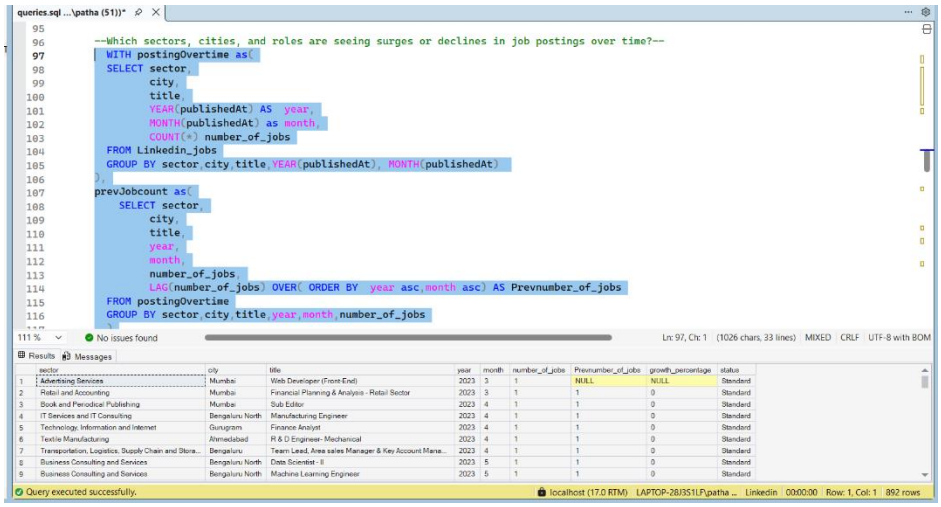
Time-series analysis of postings across sectors, cities, and roles reveals where the market is accelerating versus cooling. This is critical for strategic workforce planning and job search timing.

SQL Query Used:

```
  WITH postingOvertime as(
  SELECT sector,
         city,
         title,
         YEAR(publishedAt) AS  year,
         MONTH(publishedAt) as month,
         COUNT(*) number_of_jobs
  FROM Linkedin_jobs
  GROUP BY sector,city,title,YEAR(publishedAt), MONTH(publishedAt)
),
prevJobcount as(
    SELECT sector,
         city,
         title,
         year,
         month,
         number_of_jobs,
```

```
        LAG(number_of_jobs) OVER( ORDER BY  year asc,month asc) AS
Prevnumber_of_jobs
  FROM postingOvertime
  GROUP BY sector,city,title,year,month,number_of_jobs
  )
  SELECT sector,
        city,
        title,
        year,
        month,
        number_of_jobs,
        Prevnumber_of_jobs,
        ROUND((number_of_jobs-Prevnumber_of_jobs)*100/Prevnumber_of_jobs,2)AS
growth_percentage,
        CASE WHEN number_of_jobs>Prevnumber_of_jobs THEN 'Serge'
             WHEN number_of_jobs<Prevnumber_of_jobs THEN 'Declain'
             ELSE 'Standard' end as status
FROM prevJobcount;
```

Output / Visualization:



Q4. Experience Gaps – Are Employers Disproportionately Seeking Senior Talent?

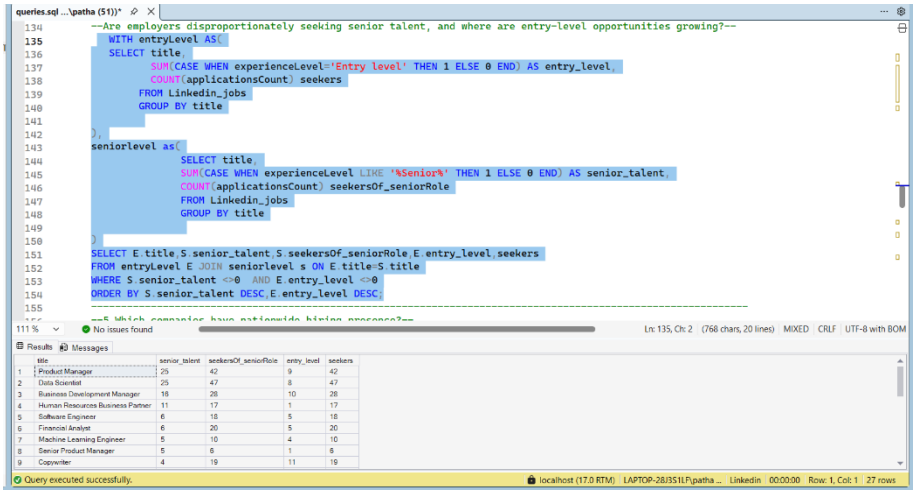
Examining the ratio of senior-level to entry-level postings by sector and city reveals potential talent pipeline imbalances — useful for policymakers, recruiters, and career advisors.

SQL Query Used:

```
SELECT sector,
       SUM(CASE WHEN experienceLevel IN ('Director','Executive','Mid-Senior level')
THEN 1 ELSE 0 END) AS senior_roles,
```

```
       SUM(CASE WHEN experienceLevel IN ('Entry level','Internship') THEN 1 ELSE 0
END) AS entry_roles,
       ROUND(SUM(CASE WHEN experienceLevel IN ('Director','Executive','Mid-Senior
level')THEN 1.0 ELSE 0 END)
       /NULLIF(SUM(CASE WHEN experienceLevel IN ('Entry level','Internship')THEN 1
ELSE 0 END),0), 2) AS senior_to_entry_ratio
FROM Linkedin_jobs
GROUP BY sector
ORDER BY senior_to_entry_ratio DESC;
```

Output / Visualization:



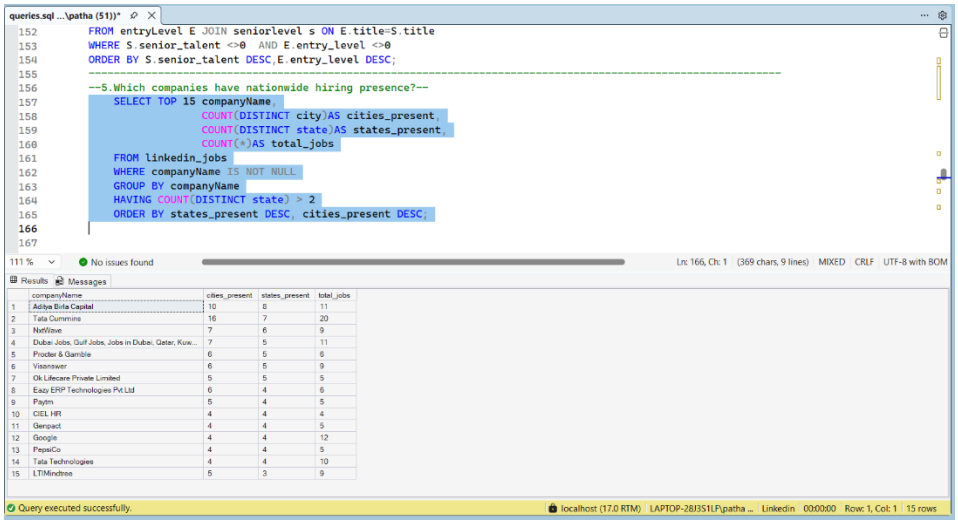
Q5. Which Companies Have Nationwide Hiring Presence?

Companies posting jobs across multiple states demonstrate a distributed workforce strategy and national brand reach — an important dimension for evaluating employer scale and remote-work culture.

SQL Query Used:

```
S    SELECT TOP 15 companyName,
                   COUNT(DISTINCT city)AS cities_present,
                   COUNT(DISTINCT state)AS states_present,
                   COUNT(*)AS total_jobs
    FROM linkedin_jobs
    WHERE companyName IS NOT NULL
    GROUP BY companyName
    HAVING COUNT(DISTINCT state) > 2
    ORDER BY states_present DESC, cities_present DESC;
```

Output / Visualization:



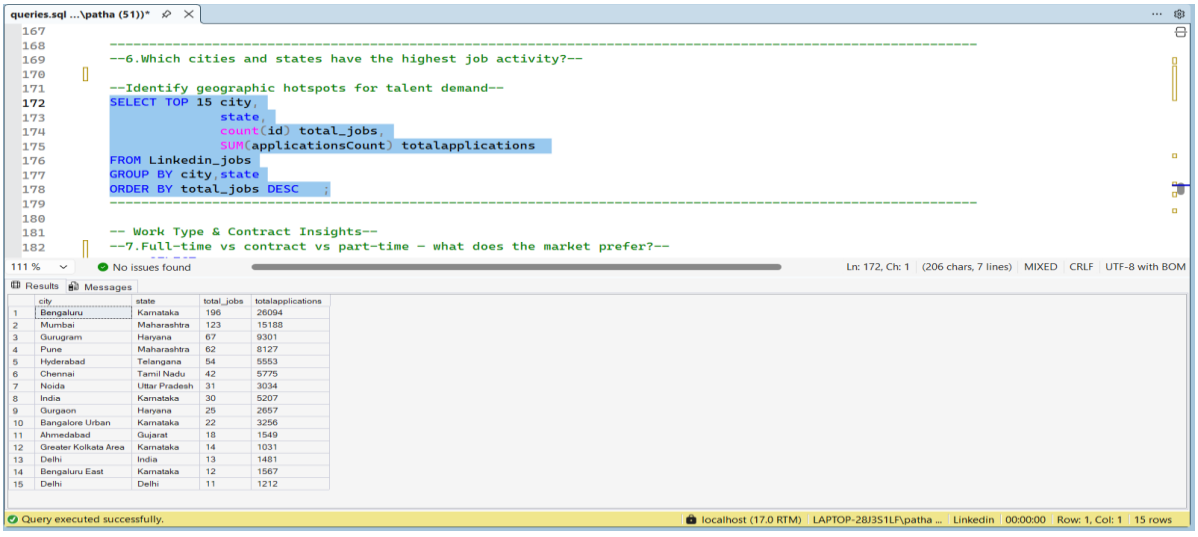
Q6. Which Cities and States Have the Highest Job Activity?

Geographic concentration of job postings identifies the top labour markets. This guides candidates on relocation decisions and reveals where employers are building out teams fastest.

SQL Query Used:

```
SELECT TOP 15 city,
              state,
              count(id) total_jobs,
              SUM(applicationsCount) totalapplications
FROM Linkedin_jobs
GROUP BY city,state
ORDER BY total_jobs DESC   ;
```

Output / Visualization:



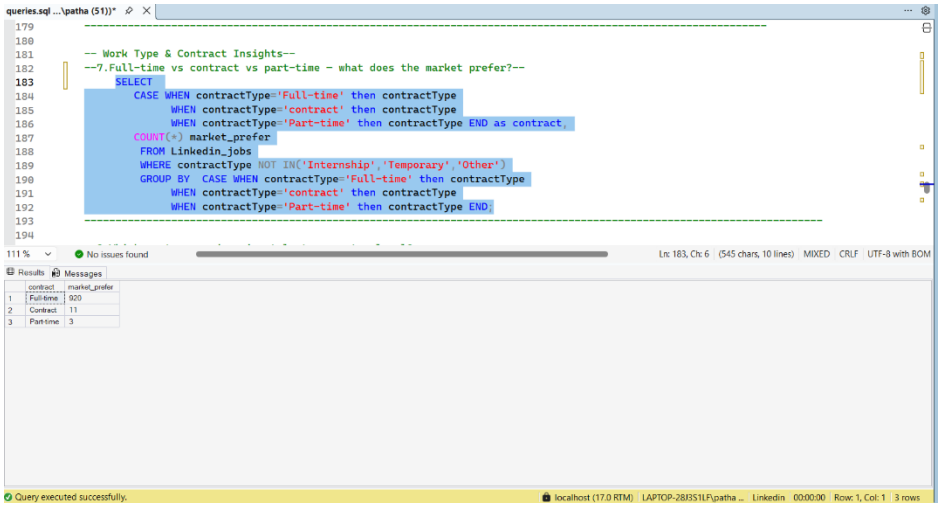
Q7. Employment Type – Full-time vs Contract vs Part-time Market Preference

Understanding the distribution of contract types reveals what employers prioritise — stable headcount growth (full-time) versus flexible capacity (contract/part-time). This shapes negotiation leverage for candidates.

SQL Query Used:

```
SELECT
   CASE WHEN contractType='Full-time' then contractType
        WHEN contractType='contract' then contractType
        WHEN contractType='Part-time' then contractType END as contract,
   COUNT(*) market_prefer
 FROM Linkedin_jobs
 WHERE contractType NOT IN('Internship','Temporary','Other')
 GROUP BY  CASE WHEN contractType='Full-time' then contractType
           WHEN contractType='contract' then contractType
           WHEN contractType='Part-time' then contractType END;
```

Output / Visualization:



Q8. Which Sectors Need Senior Talent vs Entry-Level?

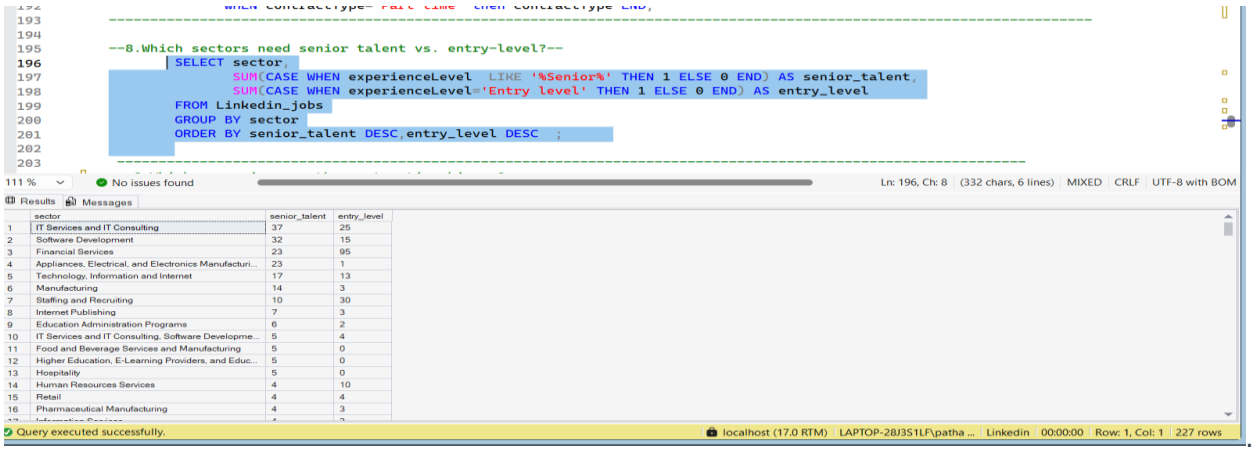
A sector-level breakdown of experience level demand allows stakeholders to understand where the talent war is focused and where fresh talent pipelines are being developed.

SQL Query Used:

```
        SELECT sector,
               SUM(CASE WHEN experienceLevel  LIKE '%Senior%' THEN 1 ELSE 0 END) AS
senior_talent,
```

```
            SUM(CASE WHEN experienceLevel='Entry level' THEN 1 ELSE 0 END) AS
entry_level
        FROM Linkedin_jobs
        GROUP BY sector
        ORDER BY senior_talent DESC,entry_level DESC  ;
```

Output / Visualization:



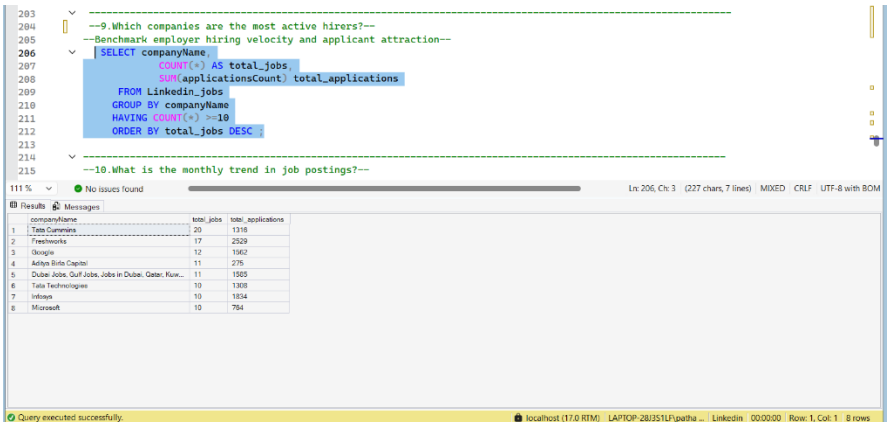
Q9. Which Companies Are the Most Active Hirers?

Combining posting volume, applicant attraction, and geographic spread gives a holistic view of the market's most dominant hiring organisations — key benchmarks for competitive intelligence.

SQL Query Used:

```
SELECT companyName,
          COUNT(*) AS total_jobs,
          SUM(applicationsCount) total_applications
    FROM Linkedin_jobs
   GROUP BY companyName
   HAVING COUNT(*) >=10
   ORDER BY total_jobs DESC ;
```

Output / Visualization:



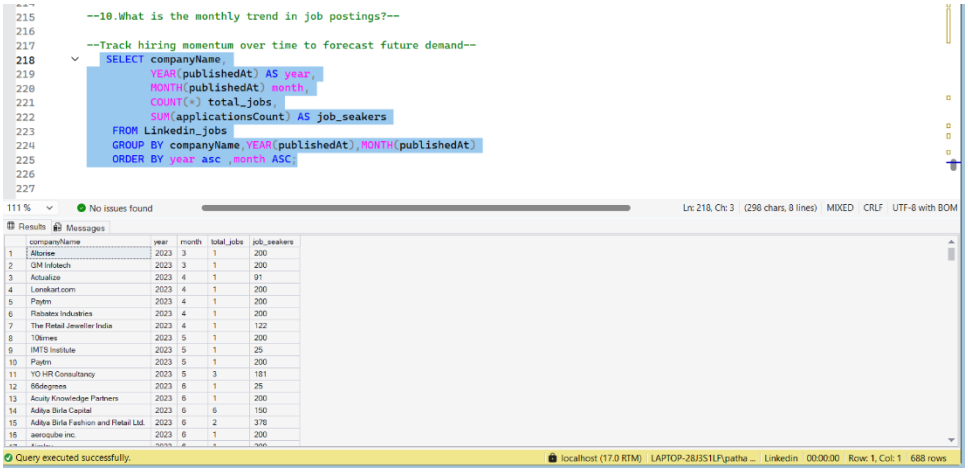
Q10. Monthly Trend in Job Postings

A time-series view of monthly posting volumes reveals the hiring calendar — when the market heats up, plateaus, or cools. This is foundational for workforce planning and candidate strategy.

SQL Query Used:

```
SELECT companyName,
       YEAR(publishedAt) AS year,
       MONTH(publishedAt) month,
       COUNT(*) total_jobs,
       SUM(applicationsCount) AS job_seakers
 FROM Linkedin_jobs
 GROUP BY companyName,YEAR(publishedAt),MONTH(publishedAt)
 ORDER BY year asc ,month ASC;
```

Output / Visualization:



Summary & Conclusions

This analysis of 949 LinkedIn job postings surfaced several high-value insights for candidates, hiring managers, and workforce strategists alike. Below is a consolidated summary of the most actionable findings.

Theme	Key Takeaway
Top Hiring City	Bengaluru (Karnataka) dominates with 196 jobs & 26,094 applications — India's #1 hiring hub
Peak Hiring Month	January leads massively with 587 postings — over 5x any other month in the dataset
Employment Type	Full-time overwhelmingly rules at 920 postings (98.3%) vs Contract (11) and Part-time (3)
Most Competed Role	Data Scientist — 6,325 total applications & DENSE_RANK #1 with 47 postings across companies
Top Sector	Financial Services leads all sectors with 131 jobs and 14,528 total applications
Employment Type	Full-time dominates; contract roles are growing, especially in tech and consulting.
Nationwide Hirers	Large enterprises operate across 5+ states, signalling distributed workforce strategies.

End of Report

For queries or collaboration: [pathanafridhkhan@gmail.com]